Comprehensive Data Source Discovery and Integration Architecture for "Fessi": A Knowledge Graph for Textile Waste Management

1. Executive Summary

The transition of the textile industry from a linear "take-make-waste" model to a circular economy represents one of the most significant data challenges in modern waste management. For "Fessi," the German university waste disposal chatbot, this challenge is acute. The system must navigate a complex matrix of material science, evolving European Union legislation, German federal mandates (KrWG), and localized disposal infrastructure. This research report provides an exhaustive analysis of the data landscape required to populate the Fessi Knowledge Graph, focusing specifically on **Textiles and Clothing**.

The research identifies that the domain is currently bifurcated between legacy, static data structures (standard lists, PDF regulations) and emerging, dynamic interoperability standards driven by the EU's Digital Product Passport (DPP) initiative. To remain viable through 2030, Fessi's architecture must ingest current "ground truth" data while establishing schema compliance for future traceability standards.

Key Strategic Findings

1. The Primacy of Molecular Identification
The release of the NIST NIR-SORT Database in 2025 fundamentally alters the capability of waste management systems to classify textiles. By providing the near-infrared spectral "fingerprints" of 64 distinct fiber types and blends, this source moves identification from reliance on often inaccurate consumer labels to scientific verification. For Fessi, this database serves as the foundational ontology for the WasteItem entity, enabling the knowledge graph to distinguish between high-value pure fibers (e.g., 100% Cotton) and difficult-to-recycle blends (e.g., Poly-cotton with >5% Elastane).1
2. The Regulatory Logic Layer
Determination of "waste" versus "product" status is not merely a physical assessment but a legal one. The LAGA Mitteilung 40 serves as the definitive German enforcement aid for this distinction. It provides the conditional logic necessary to program the chatbot's decision trees, specifically regarding "End-of-Waste" status. Integration of these rules allows the system to guide users correctly: identifying when a torn shirt must enter the Restmüll stream versus when a wearable garment can enter the Altkleidercontainer reuse stream.3
3. Infrastructure and Location Intelligence

While generic mapping services exist, OpenStreetMap (OSM) offers the most robust, community-maintained dataset for decentralized disposal infrastructure in Germany. The specific tagging schema recycling:clothes combined with amenity=recycling provides high-granularity location data. However, this must be augmented by municipal Abfallkalender APIs (e.g., Abfall+, municipal Open Data portals) to capture the temporal dimensions of waste collection—such as the schedules for Schadstoffmobile which accept contaminated textiles.4
4. Future-Proofing via the Digital Product Passport (DPP)
The research indicates that static databases will eventually be superseded by item-level digital identities. The CIRPASS consortium's reference architecture and GS1 Digital Link standards are the blueprints for this future. Fessi must adopt an ontology compatible with these JSON-LD structures now. This ensures that as brands like Nike and H&M begin rolling out DPP-enabled garments (mandated by 2027-2030), the knowledge graph can ingest this rich traceability data without architectural refactoring.6

Core Recommendations

The report recommends a phased integration strategy:

1. **Immediate Implementation:** Ingest **AVV Codes** and **LAGA M 40** rules to establish the legal framework. Use **OSM** for location services.
2. **Material Intelligence:** Integrate **NIST NIR-SORT** and **T-REX Project** guidelines to build a sophisticated recyclability engine.
3. **Traceability & Ethics:** Connect with **Open Supply Hub** and **Good On You** to provide users with transparency regarding the upstream impact of their disposal choices.

The following sections detail the strategic context and provide deep-dive profiles for 15 critical data sources, mapping their integration into the Fessi Knowledge Graph.

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2. Strategic Context & Domain Analysis

To effectively model the data sources, one must first understand the "physics" of the domain: the regulatory pressures, material complexities, and infrastructural realities of the German textile waste sector.

2.1 The Regulatory Tsunami: EU Strategy and German Law

The waste management landscape is being reshaped by the **EU Strategy for Sustainable and Circular Textiles** (2022). The critical driver for Fessi is the revision of the Waste Framework Directive (WFD), which mandates the separate collection of textile waste by January 1, 2025. This creates a binary state for waste management systems: compliance or obsolescence.

In Germany, this is transposed through the **Kreislaufwirtschaftsgesetz (KrWG)**. The distinction between *Alttextilien* (used textiles for reuse) and *Textilabfall* (textile waste) is

paramount.

- **Reuse (Wiederverwendung):** The highest tier of the waste hierarchy. Collections by charities (FairWertung members) or commercial entities aim for this.
- **Recycling (Verwertung):** Mechanical or chemical processing.
- **Disposal (Beseitigung):** Thermal recovery (incineration), which the EU strategy aims to minimize.

Implications for Fessi: The chatbot cannot simply say "Throw it away." It must act as a triage nurse for textiles. It requires data sources that define the *thresholds* between these states. If a user asks about a "paint-stained t-shirt," the system must recognize that chemical contamination (paint) overrides the material potential (cotton), shifting the classification from AVV 20 01 10 (Clothes) to potentially hazardous waste or Restmüll.[8]

2.2 The "Monstrous Hybrid" Problem: Blends and Recyclability

The core technical challenge in textile recycling—and therefore in data modeling—is the prevalence of fiber blends. The **T-REX Project** (Textile Recycling Excellence) has highlighted that while mechanical recycling of pure cotton is established, blends are problematic.

- **Polyester/Cotton:** The most common blend. Chemical separation is energy-intensive and currently rare at industrial scale.
- **Elastane (Spandex):** The "contaminant." Even 5% elastane can clog mechanical shredders and disrupt chemical recycling of polyester.[2]

Data Requirement: Fessi needs a "Recyclability Index" mapped to material composition. It is not enough to know an item is "Cotton." We must know if it is "95% Cotton / 5% Elastane" versus "100% Cotton." The **NIST NIR-SORT** database provides the spectral data to validate these distinctions, serving as the ground truth for what materials *physically exist* in the waste stream, distinct from what marketing labels might claim.[1]

2.3 The Digital Product Passport (DPP) Horizon

The **Ecodesign for Sustainable Products Regulation (ESPR)** introduces the DPP. This will essentially be a "digital twin" for every garment, accessible via a data carrier (QR code/RFID).

- **Data Fields:** Fiber composition, origin, recycled content, presence of substances of concern.
- **Architecture:** Decentralized storage, accessed via **GS1 Digital Link** resolvers.

Future-Proofing Fessi: The knowledge graph must be able to consume JSON-LD data schemas defined by the **CIRPASS** consortium. While full implementation is years away, the ontology (the definitions of "Material," "Product," "Manufacturer") must align with these emerging standards today to avoid technical debt.[6]

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3. Data Source Architecture: Material Composition & Science

This section details sources that define the physical and chemical reality of waste items. These sources populate the WasteItem and Material entities in the Knowledge Graph.

Source 1: NIST NIR-SORT Database (2025)

Source Identification

- **Name:** Near-Infrared Spectra of Origin-defined and Real-world Textiles (NIR-SORT)
- **Provider:** National Institute of Standards and Technology (NIST) (USA)
- **URL:** https://www.nist.gov/news-events/news/2025/01/nist-database-can-help-increase-recycling-textiles-and-clothing [1]

Data Format & Access

- **Format:** JCAMP-DX (Scientific standard for spectroscopy), CSV (Metadata).
- **Access Method:** Free (Public Domain). Downloadable from NIST Public Data Repository.
- **Coverage:** 64 distinct fabric types, including pure fibers and complex blends.
- **Data Quality:** Scientific "Gold Standard." Peer-reviewed government data.

Detailed Profile
The NIR-SORT database addresses the "identification gap" in automated sorting. While Fessi is a text-based chatbot, this database provides the taxonomy of reality. It lists exactly which fiber combinations appear in real-world waste streams (sourced from thrift stores), contrasting them with pristine lab samples. This allows the system to understand the degradation of materials—how a "washed 50 times" cotton shirt differs from a new one.
Integration Logic & Schema Mapping

- **Entity Mapping:** Material nodes.
- **Attributes:**
  - polymer_type: (e.g., Polyethylene Terephthalate, Cellulose).
  - blend_ratio: (e.g., 60/40).
  - condition_state: (e.g., "Pristine", "Worn", "Laundered").
- Graph Logic:
  The database allows Fessi to infer properties. If a user says "Old Gym Shirt," the graph can query the NIR-SORT metadata for "Sportswear" and find that 85% of such samples are Polyester/Elastane blends. This probability informs the disposal recommendation.
  - *Rule:* IF ItemCategory = "Sportswear" AND User_Input_Material = NULL -> Inferred_Material = "Polyester/Elastane" -> Recyclability = "Low" -> Instruction = "Check label for 'Elastane'. If present, item acts as contaminant in PET recycling."

Relevance Assessment

- **Overall Score:** 5/5 (Essential). It is the only open-source scientific baseline for post-consumer textile composition.

Source 2: ISO 3758 & ISO 2076 (via GINETEX)

Source Identification

- **Name:** ISO 2076 (Generic names of man-made fibres) & ISO 3758 (Care labelling code using symbols)
- **Provider:** ISO / GINETEX.
- **URL:** https://www.ginetex.net/gb/labelling/care-symbols.asp [11]

Data Format & Access

- **Format:** PDF Standards, Web Definitions.
- **Access Method:** Restricted (Standards are paid). Definitions viewable online.
- **Licensing:** GINETEX symbols are registered trademarks. **Commercial use of the images requires a license.**
- **Coverage:** Global (ISO).

Detailed Profile
These standards are the language of the textile label. ISO 2076 defines the allowable generic names (e.g., "Modal," "Lyocell," "Viscose" are all chemically cellulose but distinct in processing). ISO 3758 defines the care symbols.

- **Constraint:** Due to trademark restrictions, Fessi should *not* display the symbol images (SVG/PNG) without a license. Instead, it should map the *description* or *Unicode equivalent* to the logic.

Integration Logic & Schema Mapping

- **Entity Mapping:** CareInstruction nodes.
- **Attributes:**
  - symbol_id: (e.g., "washing_hand").
  - unicode_char: (e.g., U+1F9FA for Basket - verify specific symbol block support).
  - temperature_limit: (Integer).
  - process_type: (Enum: Wash, Bleach, Dry, Iron, Prof).
- Graph Logic:
  Care labels act as proxies for material durability. An item marked "Do Not Wash" often implies delicate fibers (Wool, Silk) or complex structures (Suit Jackets) that are unsuitable for standard recycling shredders.
  - *Query:* MATCH (i:WasteItem)-->(c:CareInstruction {type: 'Professional Cleaning'}) RETURN i.disposal_route -> Result: "Likely high value or complex construction.

Prioritize reuse/resale over recycling bin."

Relevance Assessment

- **Overall Score:** 4/5 (High). Essential for parsing user input about labels.

Source 3: MatWeb Material Property Data

Source Identification

- **Name:** MatWeb Material Property Database
- **Provider:** MatWeb, LLC.
- **URL:** https://www.matweb.com/ [13]

Data Format & Access

- **Format:** Web (HTML), Premium Export (Excel/XML).
- **Access Method:** Freemium.
- **Coverage:** Technical properties of polymers (tensile strength, melting point, density).

Detailed Profile
While not specific to "fashion," MatWeb provides the physics of the underlying polymers. For technical textiles (e.g., university lab coats, protective gear), knowing the melting point of Aramid vs. Polyester is critical for determining thermal recycling compatibility.
Integration Logic & Schema Mapping

- **Entity Mapping:** Polymer nodes linked to Material.
- **Attributes:** melting_point_celsius, density_g_cm3, biodegradability_status.
- **Integration:** Use as a lookup table for the Material entity. When Fessi encounters "Nylon 6.6", it pulls the thermal properties from MatWeb to determine if it can be co-processed with "Nylon 6" (often no).

Relevance Assessment

- **Overall Score:** 3/5 (Supporting). Provides depth for technical/scientific waste queries typical in a university setting.

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4. Data Source Architecture: Logic, Rules & Recyclability

This layer is the "Brain" of Fessi. It translates the physical reality of an item (from Layer 1) into a disposal decision based on legal and technical constraints.

Source 4: LAGA Mitteilung 40 (German Enforcement Aid)

Source Identification

- **Name:** Vollzugshilfe zur Vermeidung sowie zur Erfassung, Sortierung und Verwertung von Alttextilien (M 40)
- **Provider:** Bund/Länder-Arbeitsgemeinschaft Abfall (LAGA).
- **URL:** https://www.laga-online.de/documents/laga-m40-vollzugshilfe-zur-vermeidung-sowie-zur-erfassung,-sortierung-und-verwertung-von-alttextilien_1685020405.pdf [3]

Data Format & Access

- **Format:** PDF (Unstructured text).
- **Access Method:** Free.
- **Coverage:** Germany (Federal).

Detailed Profile
This document is the "Code of Law" for textile waste. It explicitly defines the "End-of-Waste" criteria.

- **Key Concept:** Textiles are only "prepared for reuse" (and thus cease to be waste) if they are sorted, checked, and a market exists.
- **Sorting Criteria:** It defines "Teilsortierung" (Partial) vs "Vollsortierung" (Full).
- **Hazard Markers:** It explicitly states that moldy/wet textiles belong in Restmüll or incineration, not textile banks.

Integration Logic & Schema Mapping

- **Entity Mapping:** DisposalRule and Condition.
- **Graph Logic (Decision Tree):**
  - *Node:* Condition:Wet
  - *Edge:* VIOLATES_RULE
  - *Target:* Rule:LAGA_M40_Dry_Storage
  - *Result:* Disposal_Action = "Dispose in Restmüll immediately. Do not place in Altkleidercontainer."
- **Technical Note:** This requires manual extraction (NLP or human review) to convert PDF text into graph rules (SHACL shapes or simple logic nodes).

Relevance Assessment

- **Overall Score:** 5/5 (Critical). Without this, the chatbot gives illegal or operationally incorrect advice for Germany.

Source 5: T-REX Project Technical Guidance (2025)

Source Identification

- **Name:** T-REX Technical Guidance for Recyclable Garments

- **Provider:** T-REX Project (EU Horizon 2020 Consortium).
- **URL:**(https://circulartextiles.aalto.fi/trexproject/wp-content/uploads/2025/06/T-REX-Technical-Guidance-For-Recyclable-Garments.pdf) [2]

Data Format & Access

- **Format:** PDF.
- **Access Method:** Free.
- **Coverage:** EU. Focused on Polyester, Polyamide 6, Cotton.

Detailed Profile
While LAGA handles the legal side, T-REX handles the technical recyclability. It introduces a "Traffic Light" system for materials.

- **Red (Disrupter):** Material makes the whole garment unrecyclable (e.g., >5% Elastane in Polyester).
- **Yellow (Compatible):** Accepted within thresholds.
- **Green (Preferred):** Mono-material.

Integration Logic & Schema Mapping

- **Entity Mapping:** RecyclabilityConstraint.
- **Attributes:** threshold_percentage, disruptor_material, base_material.
- **Graph Logic:**
  - *Query:* MATCH (m:Material {name: 'Elastane'})-->(i:WasteItem) WHERE r.percentage > 5 RETURN 'Disruptor'
  - *User Output:* "This item has too much stretch (Elastane) for standard textile recycling. If not wearable, it will likely be downcycled or incinerated."

Relevance Assessment

- **Overall Score:** 5/5 (High). Provides the nuance required for a "smart" chatbot.

Source 6: Textile Exchange Preferred Fiber & Materials Matrix

Source Identification

- **Name:** Fiber and Materials Matrix (PFMM)
- **Provider:** Textile Exchange.
- **URL:** https://textileexchange.org/pfmm/ [16]

Data Format & Access

- **Format:** Interactive Web Tool / Reports.
- **Access Method:** Free / Registration.
- **Coverage:** Global sustainability metrics (Water, Energy, Chemical use).

Detailed Profile
This source adds the "Sustainability" layer. It helps Fessi answer questions like "Is bamboo better than cotton?" by providing standardized impact assessments. It categorizes fibers into "Preferred" vs. "Conventional."

Integration Logic & Schema Mapping

- **Entity Mapping:** SustainabilityMetric linked to Material.
- **Attributes:** water_impact_score, carbon_footprint_relative.
- **Application:** Educational feedback. "You are disposing of a Conventional Cotton shirt. Did you know Organic Cotton uses significantly less water?"

Relevance Assessment

- **Overall Score:** 3/5 (Contextual). Good for user engagement, less critical for strict disposal routing.

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5. Data Source Architecture: Infrastructure & Supply Chain

This layer provides the geospatial and organizational data to answer "Where do I go?"

Source 7: OpenStreetMap (OSM)

Source Identification

- **Name:** OpenStreetMap
- **Provider:** OSM Foundation.
- **URL:** https://overpass-turbo.eu/ [18]

Data Format & Access

- **Format:** XML / JSON / GeoJSON (via Overpass API).
- **Access Method:** Free (ODbL).
- **Coverage:** Global, excellent density in Germany.

Detailed Profile
OSM is the definitive source for decentralized infrastructure. Users map individual containers, often faster than municipalities update their lists.

- **Tags of Interest:**
  - amenity=recycling
  - recycling:clothes=yes
  - recycling:shoes=yes
  - operator=* (e.g., "DRK", "Malteser", "Private Commercial").

Integration Logic & Schema Mapping

- **Entity Mapping:** Container, Location.
- Technical Implementation:
  Use an Overpass QL query to fetch nodes within the university city bounding box.
  C
  ```
  [out:json];
  area["name"="Heidelberg"]->.searchArea; // Example University City
  (
    node["amenity"="recycling"]["recycling:clothes"="yes"](area.searchArea);
    node["social_facility"="clothing_bank"](area.searchArea);
  );
  out body;
  >;
  out skel qt;
  ```

- **Data Cleaning:** Filter out nodes with access=private.

Relevance Assessment

- **Overall Score:** 5/5 (Essential).

Source 8: Local Municipal "Abfallkalender" APIs

Source Identification

- **Name:** My-Abfallkalender / Abfall+ / City Open Data Portals
- **Provider:** Local Waste Management Utilities (e.g., AWB, BSR).
- **URL Examples:**
  - Würzburg API: https://opendata.wuerzburg.de/explore/dataset/abfallkalender-wuerzburg/api/ [5]
  - Mannheim API: https://mannheim.opendatasoft.com/explore/dataset/150824-altkleidercontainer-mannheim/api/ [20]

Data Format & Access

- **Format:** REST API (JSON), ICS (iCal), CSV.
- **Access Method:** Free (Open Data Germany).
- **Coverage:** City-specific.

Detailed Profile
These are the "Official" sources. They contain data OSM might miss, such as the schedule for the Schadstoffmobil (Hazardous Waste Mobile) or temporary closures of Wertstoffhöfe (Recycling Centers).

Integration Logic & Schema Mapping

- **Entity Mapping:** Facility with Schedule.
- **Logic:**
  - IF Item = "Contaminated Textile (Paint/Oil)" -> Find next Schadstoffmobil date from API -> Return Date & Location.

Relevance Assessment

- **Overall Score:** 5/5 (Essential). Legal authority on disposal.

Source 9: FairWertung & BVSE (Trusted Collectors)

Source Identification

- **Name:** FairWertung Standortsuche / BVSE Mitgliedersuche
- **Provider:** Dachverband FairWertung e.V. / BVSE.
- **URL:** https://fairwertung.de/ [21], https://www.bvse.de/ [22]

Detailed Profile
In Germany, illegal container placement is a problem. FairWertung represents charitable, ethical collectors. BVSE represents certified commercial recyclers.

- **Data Format:** Mostly HTML map interfaces (Web Scraping required).
- **Value:** Trust. Fessi can flag containers as "Ethically Certified" if they appear in these lists.

Integration Logic

- **Entity Property:** TrustScore or OrganizationType.
- **Logic:** Sort disposal results to prioritize FairWertung locations over generic/unknown containers.

Relevance Assessment

- **Overall Score:** 3/5 (High trust, low technical accessibility).

Source 10: Open Supply Hub (OAR)

Source Identification

- **Name:** Open Supply Hub (formerly Open Apparel Registry)
- **Provider:** Non-profit.
- **URL:** https://opensupplyhub.org/api/docs/ [23]

Data Format & Access

- **Format:** REST API.
- **Access Method:** Free (Registration required).

Detailed Profile
Maps global apparel facilities. While it doesn't list products, it links Brands -> Suppliers.

- **Use Case:** If Fessi incorporates a brand transparency feature ("How sustainable is the brand of this shirt?"), OAR provides the supply chain map.

Relevance Assessment

- **Overall Score:** 4/5 (Supply Chain Context).

Source 11: Good On You (Brand Ratings)

Source Identification

- **Name:** Good On You API
- **Provider:** Good On You (Social Enterprise).
- **URL:** https://partnerships.goodonyou.eco/solutions/publishers [25]

Data Format & Access

- **Format:** JSON API.
- **Access Method:** Paid / Commercial Partnership required.
- **Coverage:** Thousands of brands rated on People, Planet, Animals.

Detailed Profile
The leading source for brand sustainability ratings.

- **Integration:** Can be used to annotate the Brand entity in the graph. When a user discards a "Shein" item, Fessi could (optionally) display its low durability rating, reinforcing the need for proper disposal vs. resale.

Relevance Assessment

- **Overall Score:** 3/5 (High value, high barrier to entry due to cost).

Source 12: CIRPASS (Digital Product Passport)

Source Identification

- **Name:** CIRPASS (Collaborative Initiative for a Standards-based Digital Product Passport)
- **Provider:** EU Consortium.
- **URL:** https://cirpassproject.eu/ [6]

Detailed Profile
CIRPASS is defining the schema for the future. It doesn't hold data itself but defines how data must be structured.

- **Key Fields:** unique_product_identifier, recyclability_information, substances_of_concern.

Integration: Fessi's graph ontology should use CIRPASS class names (e.g., dpp:RecycledContent) to ensure that when DPP data becomes available (2027+), the graph is ready to accept it.

Relevance Assessment

- **Overall Score:** 5/5 (Architecture Definition).

Source 13: GS1 Digital Link & Barcode Lookup

Source Identification

- **Name:** GS1 Digital Link / Barcode Lookup
- **Provider:** GS1 / Commercial Providers (e.g., Barcode Lookup).
- **URL:** https://www.gs1.org/standards/gs1-digital-link [7], https://www.barcodelookup.com/api [28]

Detailed Profile
The mechanism for the DPP. A barcode scan resolves to a URL https://id.brand.com/01/{GTIN}.

- **Integration:** Fessi should implement a "Scan Barcode" feature. The backend will query Barcode Lookup databases to identify the item name/brand, then query the internal Graph for disposal logic.

Relevance Assessment

- **Overall Score:** 4/5 (Identification).

Source 14: Fashion Revolution Transparency Index

Source Identification

- **Name:** Fashion Transparency Index
- **Provider:** Fashion Revolution (NGO).
- **URL:** https://www.fashionrevolution.org/
- **Format:** PDF Report / CSV (sometimes released).

Detailed Profile
Ranks top 250 brands on disclosure. Use this to flag brands that should have take-back programs, prompting Fessi to suggest "Check Brand Website for Take-Back" for high-ranking brands.
Relevance Assessment

- **Overall Score:** 2/5 (Static, high-level).

Source 15: Wikidata (Textile Properties)

Source Identification

- **Name:** Wikidata
- **Provider:** Wikimedia Foundation.
- **URL:** https://www.wikidata.org/
- **Format:** SPARQL Endpoint.
- **Access Method:** Free.

Detailed Profile
A structured knowledge base.

- **Query:** Get all instances of "synthetic fiber" with "recycling code."
- **Value:** Multilingual synonyms (English "Cotton" = German "Baumwolle"). Essential for the chatbot's NLP layer to understand German user queries.

Relevance Assessment

- **Overall Score:** 4/5 (NLP support).

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6. Integration Recommendations & Schema Design

To transform these disparate sources into a cohesive "Fessi" Knowledge Graph, a specific ETL (Extract, Transform, Load) and schema strategy is required.

6.1 Phase 1: Core Legal & Location Logic (The "Must Haves")

- **Logic:** Ingest **LAGA M 40** rules manually. Define the Condition nodes (Clean/Dirty, Whole/Torn).
- **Infrastructure:** Set up the **Overpass API** pipeline to fetch OSM data for the university city. Filter for recycling:clothes.
- **Vocabulary:** Import **Wikidata** synonyms for textile types to bootstrap the NLP entity recognition.

6.2 Phase 2: Material Intelligence (The "Brain")

- **Spectroscopy Integration:** Ingest the **NIST NIR-SORT** metadata. Map specific "Real World" samples to WasteItem prototypes.
- **Recyclability Logic:** Encode the **T-REX** "Traffic Light" system as graph edges.
  - Material(Polyester) - -> Process(Mechanical_Recycling)
  - Material(Polyester) - -> Material(Elastane)
- **Label Parsing:** Map **ISO 3758** text descriptions to disposal constraints.

6.3 Phase 3: Future-Proofing (DPP & Barcodes)

- **Schema Alignment:** Update the graph ontology to match **CIRPASS** draft specs. Ensure fields like carbon_footprint and supply_chain_origin exist, even if empty.
- **Barcode Resolver:** Implement a connector to **GS1 Digital Link** resolvers. When a user provides a URL or barcode, the system attempts to fetch the JSON-LD passport data directly.

6.4 Proposed Graph Schema (Mermaid)

Code snippet

```
graph TD
    %% Core Item Definition
    Item -->|HAS_COMPOSITION| Comp[MaterialComposition]
    Item -->|IDENTIFIED_BY| GTIN
    Item -->|HAS_CONDITION| Cond[Condition]

    %% Material Science Layer (NIST/T-REX)
    Comp -->|CONTAINS| Mat[Material]
    Mat -->|DEFINED_BY| NIST
    Mat -->|HAS_PROPERTY| Prop

    %% Logic Layer (LAGA/T-REX)
    Comp -->|RECYCLABILITY_STATUS| RStatus
    RStatus -->|DETERMINED_BY| Trex

    Item -->|GOVERNED_BY| Law

    %% Infrastructure Layer (OSM/Municipal)
    Item -->|DISPOSE_AT| Loc[Location]
    Loc -->|IS_A| Cont[Container]
    Loc -->|IS_A| Fac

    Cont -->|LOCATED_VIA| OSM
    Fac -->|SCHEDULE_VIA| API[Municipal_API]

    %% Brand & Ethics Layer
    Item -->|PRODUCED_BY| Brand
    Brand -->|RATED_BY| GoodOnYou
    Brand -->|MAPPED_IN| OAR
```

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7. Gaps Identified

Despite the richness of the identified sources, the following gaps persist:

1. **Real-Time Brand Take-Back Inventory:** While we know H&M *has* a program, there is no public API to query "Does the H&M at Heidelberg Hauptstraße accept torn shoes *today*?" This data remains unstructured on brand websites.
    - *Mitigation:* Create manual "Brand Program" nodes for major retailers near the university, updated quarterly.
2. **Container Capacity (Fill Levels):** OSM provides location, but not status. Smart bins exist (e.g., measuring fill level via IoT), but this data is proprietary to waste management firms (Remondis, Alba) and rarely shared via Open Data.
    - *Mitigation:* Rely on user feedback ("Bin Full") to temporarily flag nodes in the graph.
3. **Chemical Additive Transparency:** NIST identifies the *fiber*, but identifying specific dyes or finishes (which might render an item hazardous) generally requires lab testing. The DPP aims to solve this with the "Substances of Concern" field, but data population will be slow.
    - *Mitigation:* Implement conservative logic. If an item is flagged as "treated" or "industrial," default to the Restmüll or Hazardous stream.

8. Conclusion

Building "Fessi" requires a sophisticated data orchestration strategy that bridges the gap between scientific precision and consumer reality. By anchoring the system in the **NIST NIR-SORT** database for material truth, **LAGA M 40** for legal compliance, and **OpenStreetMap** for location services, the university can deploy a robust, compliant tool. Crucially, by adopting the **CIRPASS** and **GS1** architectures now, Fessi will evolve from a simple disposal chatbot into a gateway for the circular economy, ready to interpret the Digital Product Passports of the future. The recommended priority is to establish the legal/location baseline first, then layer in the complex material science data to refine sorting accuracy.

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